Single Domain Generalization Method for Remote Sensing Image Segmentation via Category Consistency on Domain Randomization

Chenbin Liang, Weibin Li, Yunyun Dong, and Wenlin Fu

***Abstract*— Single domain generalization (SDG) is a more realistic setting than domain generalization (DG) and domain adaptation (DA). It aims to train a domain-agnostic model in the presence of a single source domain to perform well on arbitrary unseen target domains. To facilitate the practical deployment of remote sensing image segmentation in the real world, we propose a novel SDG method, termed category consistency on domain randomization (CCDR). To expand the coverage of a single source domain, CCDR implements a simple yet effective data generation module to perform domain randomization of texture and style information, since texture divergences from different geographical environments or phenological periods, and style divergences from different illumination or weather conditions, are the main causes of the domain shift in remote sensing. In addition to emphasizing inter and intraclass relationships within each domain via multidomain supervised learning, CCDR further draws inspiration from the triple loss to enhance the semantic correlation across the source domain and the generated auxiliary domain, which makes the segmentation model more sensitive to class discriminative information and better adapted to unseen target domains. In comparison to other state-of-the-art SDG methods, CCDR can synthesize the more effective auxiliary domain, and perform more reliable classification on the unseen target domains without the sophisticated training pipeline and cumbersome data generation process. The experimental results on two public remote sensing datasets demonstrate that CCDR has remarkable advantages in remote sensing image segmentation tasks. And the code is publicly available at: https://github.com/LCB1970/CCDR.**

***Index Terms*— Domain expansion, single domain generalization (SDG), style randomization, texture randomization, triplet loss.**

I. Introduction

FOR exploiting the potential value and prompting interdisciplinary applications of remote sensing big data, pixel-level image interpretation has always been a topic of intense scrutiny. Due to the high-precision performance and low dependence on expert knowledge, deep neural networks (DNNs), including convolutional neural networks (CNNs), attention mechanisms, and graph convolutional networks (GCNs), have recently drawn a surge of interest from remote sensing image segmentation tasks in recent years [1], [2], [3], [4], [5], [6]. And the independent and identically distributed (i.i.d.) assumption, which posits that the training data (source domain) and testing data (target domain) are drawn from similar distributions [7], is the cornerstone underlying the remarkable success of DNN-based image segmentation models. However, the distribution difference between the source and target domains, called domain shift, frequently occurs in the real world and easily brings about severe performance deterioration. It is impractical to gather enough data covering all possible shifts for model retraining, especially in the remote sensing field.

To address this problem, several appealing solutions have been developed, such as domain adaptation (DA) and domain generalization (DG). DA aims to learn a classifier for the label-scarce target domain with the aid of the label-sufficient source domain, which either employs generation models to simulate the domain shift between the source and target domains [8], [9], [10], [11], or develops some cost functions to narrow the distribution differences between the two domains in the data, feature, or decision space [12], [13], [14], [15]. In spite of the remarkable accomplishments of DA, it can only directionally transfer the model to a specific target domain where data distribution is available. Therefore, DG further employs multisource domains to empower the model with the ability to handle arbitrary unseen target domains, resorting to techniques such as representation learning [16], [17], feature disentanglement [18], [19], [20], adversarial attack [21], and meta-learning [22], [23], [24]. However, well-annotated data collection budgets significantly hinder the practical deployment of DG methods. And single DG (SDG), a more realistic setting, consequently comes into being. Therefore, considering the practical requirements in the real world, this article aims to develop a novel and effective SDG method for remote sensing image segmentation.

In order to learn domain-invariant information, it is inevitable to perform domain expansion in most SDG methods, due to the lack of diverse source domains at training time. Some research employed adversarial data augmentation (ADA)

Manuscript received 4 December 2023; revised 31 January 2024 and 4 March 2024; accepted 16 March 2024. Date of publication 20 March 2024; date of current version 2 April 2024. This work was supported in part by the China Postdoctoral Science Foundation under Grant 2023M742729 and in part by the National Natural Science Foundation of China under Grant 62001275 and Grant 42301524. *(Corresponding author: Wenlin Fu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3379669

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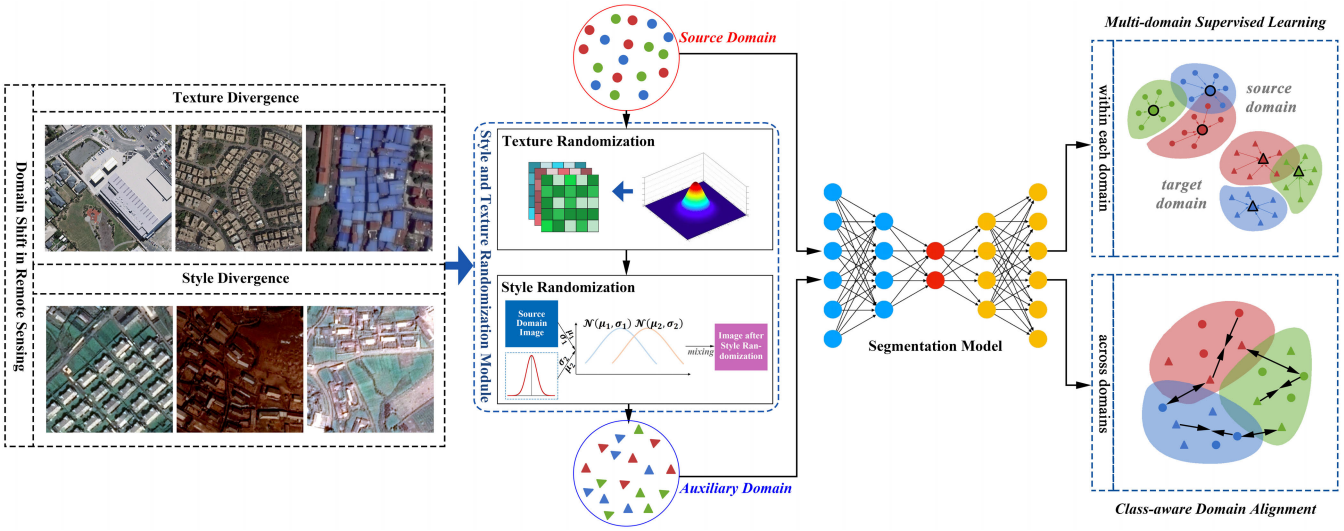


Fig. 1. Conceptual overview of CCDR. Due to the domain shift in remote sensing mainly from texture and style divergences, CCDR performs texture and style randomization for simple yet effective auxiliary domain generation. Besides, to further improve the reliability of classification in arbitrary unseen target domains, CCDR not only confirms class boundaries within each domain via multidomain supervised learning but also emphasizes the semantic correlation across domains via class-aware domain alignment.

[25], [26], [27] for domain expansion, which diversifies the source domain via confusing the classifier or maximizing its entropy. Furthermore, to synthesize more diverse changes rather than simple perturbations, other literature [28], [29], [30] utilizes additional data for stylization, or sets multiple generators for domain expansion from various views.

However, there are numerous challenges to overcome when developing an effective SDG method for remote sensing image segmentation. First, domain expansion should fully consider the complexity and particularity of domain shift problems in remote sensing, which not only include style differences caused by different illumination or weather conditions but also texture differences brought by different geographical environments or phenological periods. In other words, the SDG method for remote sensing image segmentation should ensure the **effectiveness** of the generated auxiliary domains. Second, considering the poor interclass separability and intraclass similarity in remote sensing images, as well as the presence of some low-quality synthetic data that may impair the extraction of class discriminative information, the SDG method for remote sensing image segmentation should guarantee the **reliability** of classification. Third, given the high-efficiency requirement of practical scenarios, as well as the inherently time-consuming nature of pixel-level segmentation tasks, the SDG method for remote sensing image segmentation should not impose too much burden on model parameters and training costs. In simple terms, it is crucial to maintain algorithmic **simplicity**.

At present, numerous studies on SDG have emerged, but they are unable to effectively confront the above three challenges. Some existing SDG methods [31], [32], [33] can be implemented in a simple way, but perform poorly on the unseen target domains, and cannot effectively generate diverse auxiliary domains to simulate the complex domain shift in remote sensing. Other existing SDG methods [26], [28], [29] overcome the aforementioned drawbacks using some technologies, such as adversarial learning and meta-learning, but generally require the complicated process of training or generation. In this article, we propose a novel single-DG method for remote sensing image segmentation, termed category consistency on domain randomization (CCDR), as shown in Fig. 1. In comparison to the state-of-the-art SDG methods, CCDR can offer superior performance on effectiveness, reliability, and simplicity in remote sensing image segmentation tasks. First, according to the particularity of domain shift in remote sensing, CCDR devises the texture and style randomization module (TSRM) for domain expansion, which is more **effective** than the domain expansion approaches of other SDG methods in simulating texture and style divergences in remote sensing. Second, in addition to supervised learning using the source domain and the auxiliary domain generated by TSRM, CCDR imposes the category consistency constraint via mining the class-aware triplet relationship among cross-domain samples. Combined with the diverse distributions from the auxiliary domain generated by TSRM, CCDR can perform more **reliable** classification in the unseen target domains. Third, CCDR does not involve the cumbersome backpropagation process as adversarial-based SDG methods, its auxiliary domain generation module, namely, TSRM, is an entirely stochastic process that does not necessitate any trainable parameters, and its implementation is more **simple** than that of most existing SDG methods.

In the following, the Wuhan University (WHU) and Online Map (OM) datasets for remote sensing image segmentation, which, respectively, contain three and six subdatasets collected under different imaging conditions, are employed for experimental studies. The superiority of CCDR in remote sensing image segmentation tasks is examined via comparison with other state-of-the-art SDG methods. The specific role of each component within CCDR is carefully investigated through ablation experiments. Furthermore, we also investigate the influence of crucial hyperparameters, explore the scalability of

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CCDR in different segmentation models, and conduct in-depth analysis through data and feature visualization. Overall, the main contributions can be summarized as follows.

1) Implement a novel domain expansion module, termed the TSRM, which contains multiple random convolution (RandConv) layers and a novel normalization layer, called random instance normalization (RandIN). Experimental results state that TSRM can perform effective auxiliary domain generation without the cumbersome generation process.
2) Develop a novel cross-domain constraint by mining the class-aware triplet relationship among cross-domain samples. Experimental results indicate that this constraint can effectively underscore semantic correlations across domains, and enhance classification performance in unseen target domains.
3) Propose a novel SDG method for remote sensing image segmentation, termed CCDR. Compared with other state-of-the-art SDG methods, CCDR has significant advantages in the **effectiveness** of domain expansion, the **reliability** of classification in the unseen target domains, and the **simplicity** of the training pipeline. The experimental results on the two public datasets demonstrate that CCDR has superior performance in remote sensing image segmentation tasks.

II. Related Work

A. DA for Remote Sensing Image Segmentation

The most favored strategy in DA for remote sensing image segmentation is to train one or several domain classifiers via adversarial loss from different perspectives. Liu et al. [34] proposed a bispace adversarial learning strategy that aligns feature distributions in the image and wavelet domains through adversarial learning domain classifiers in both feature and output spaces. In order to improve the effectiveness of domain alignment, Yan et al. [35] devised the triplet adversarial DA through the domain similarity discriminator rather than the domain classifier. After that, to emphasize class discriminative information in remote sensing image segmentation tasks, some researchers imposed class-aware constraints into adversarial-based DA. Chen et al. [36] performed semantic-level feature alignment via adversarial learning, and introduced the attention mechanism to pay more attention to category-level unaligned features. Zhao et al. [37] extracted domain-invariant features in an adversarial manner, and further transferred semantic relationship knowledge via Jensen–Shannon (JS) divergence.

Besides, the generative adversarial network (GAN) [38] and its variants [39], [40], [41] have also been introduced to address domain shift problems in remote sensing image segmentation tasks. Zhang et al. [42] implemented interdomain adaptation using GAN, and performed intradomain adaptation via an easy-to-hard self-training strategy. Benjdira et al. [43] and Shi et al. [44] both transferred the source domain to the target domain via CycleGAN [39] for model training, and the latter further performed feature alignment between the transferred source domain and target domain via adversarial loss. To simplify the training pipeline, Tasar et al. [45] constructed a simple yet effective image translation model by replacing the generator of traditional GAN with several matrix operations, and then adopted it to transfer the source domain to the target domain for training the segmentation model. Afterward, in order to further improve the DA effect, Li et al. [46] not only imposed the transfer invariant constraint via DualGAN, but also employed the pseudo-label constraint based on self-training, and the rotation consistency constraint inspired by self-supervised learning.

The success of these DA methods for remote sensing image segmentation is encouraging. However, these methods are deployed under the assumption that the data distribution in the target domain is accessible. Furthermore, most methods suffer from training instabilities, as a result of using adversarial learning. Therefore, this article focuses on a more realistic setting known as SDG to tackle domain shifts in remote sensing image segmentation tasks and takes full account of data complexity and algorithm simplicity.

B. Domain Generalization

DG aims to generalize the classifier from multiple source domains to the unseen target domain, and there are three prevalent approaches: learning domain-invariant feature representation [16], [17], [18], [19], [20], generating diverse samples for generalization [21], [47], [48], and exploiting the general learning strategy to promote the generalization capability [22], [23], [24]. As a special case of DG, SDG is a more challenging and realistic paradigm and has recently gained growing attention from the community, where only one source domain is available during training and the model is evaluated on multiple unseen target domains.

The key to most SDG methods is how to generate auxiliary domains with sufficient diversity and informativeness to expand the coverage of the source domain. Inspired by the development of adversarial attacks [49], ADA [25] converts SDG to the worst case problem, and appends adversarial examples to the source data at each iteration. After that, to make improvements, meta-learning-based adversarial domain augmentation (M-ADA) [26] not only uses a Wasserstein auto-encoder (WAE) to relax the worst case constraint but also follows the meta-learning scheme. Besides, some literature introduced GAN-based generation networks into SDG. Yue et al. [29] randomly drew k real-life categories from ImageNet [50] to assist in stylizing the source domain with the image-to-image translation network, and further enforced pyramid consistency across different stylized images and within an image. Progressive domain expansion network (PDEN) [28] stacks 20 generators progressively for domain expansion from various views, and further extracts domain-invariant features through contrastive learning. Afterward, other studies suggested SDG methods with low algorithm complexity by avoiding intricate generation processes. Hendrycks et al. [31] employed simple augmentation operations in concert with a consistency loss to improve the robustness and uncertainty estimates of image classifiers. Zhou et al. [32] proposed a plug-and-play module inserted between CNN layers, termed MixStyle, which mixes

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the feature statistics of training samples without the need to explicitly generate an image of a new style. Xu et al. [33] devised RandConv for simple yet effective domain expansion, and adopted Kullback–Leibler (KL) divergence to encourage consistency of network predictions.

However, the existing SDG methods cannot achieve favorable effectiveness-reliability-simplicity tradeoffs in remote sensing image segmentation tasks. Some SDG methods do not require high algorithm complexity, but they cannot effectively generate diverse auxiliary domains and perform reliable classification in the unseen target domains. Besides, although other SDG methods have obvious advantages in auxiliary domain generation and cross-domain classification, they generally require the cumbersome process of training or generation, hampering their practical deployment in remote sensing. Therefore, in order to overcome the drawbacks of the existing SDG methods and better balance effectiveness, reliability, and simplicity, our SDG method carefully analyzes the main causes of domain shift in remote sensing to implement a simple yet effective auxiliary domain generation and imposes class-aware domain alignment to further enhance the classification ability of the segmentation model in the unseen target domains.

III. Methodology

A. Overview

Our goal is to develop a remote sensing image segmentation model that can be well applied to real-world applications where domain gaps exist. Hence, we suggest a novel single-DG method, termed CCDR, as shown in Fig. 2. Specifically, given a single source domain $\mathcal{D}^S = \{(x_i^s, y_i^s)\}_{i=1}^N$, where $x_i^s \in \mathbb{R}^{H\times W\times 3}$, $y_i^s \in \mathbb{R}^{H\times W\times N_c}$, N is the number of labeled source samples, and N_c is the number of categories. First of all, CCDR performs domain randomization regarding texture and style information to generate the auxiliary domain $\mathcal{D}^A = \{(x_i^a, y_i^a)\}_{i=1}^N$ with diverse data distributions in each training epoch, where $x_i^a \in \mathbb{R}^{H\times W\times 3}$ and $y_i^a \in \mathbb{R}^{H\times W\times N_c}$. Afterward, CCDR employs $\mathcal{D}^S$ and $\mathcal{D}^A$ for multidomain supervised learning and class-aware domain alignment. The former facilitates learning domain-invariant representations by exposing the segmentation model to more domain variations. The latter is useful to enlarge the distance between features from different classes and different domains while aggregating features from the same class but different domains.

B. Auxiliary Domain Generation

To ensure the effectiveness and simplicity of domain expansion, we propose a novel auxiliary domain generation module, termed TSRM, which can effectively wrestle with complex domain shifts in remote sensing without additional training and huge computational overhead.

For texture divergences, inspired by previous filter-bank-based texture models, Xu et al. [33] employed a single convolution layer with randomly sampled weights, dubbed RandConv, to augment images with random texture. However, this augmentation approach cannot generate diverse enough images to simulate the complex domain shift in remote sensing images.

Therefore, in order to make the segmentation model robust to diverse, multiscale, and varying-degree texture divergences in remote sensing, TSRM applies multilayer RandConv (Multi_RC) to source domain data for more effective texture modifications. In Multi_RC, the number of layers, as well as the kernel size and weights of each layer, are initialized differently for each mini-batch during training. Given the input x, the process of texture randomization with Multi_RC can be formulated as follows:

$$\text{Multi_RC}(x, L) = \begin{cases} \text{Multi_RC}(x, L-1) * \Theta_L, & L > 0 \\ x, & L = 0 \end{cases} \tag{1}$$

where $*$ is the convolution operation. L is the number of layers uniformly sampled from $\{1, 2, \ldots, L_{\max}\}$, and $L_{\max}$ is a hyperparameter. $\Theta_L \in \mathbb{R}^{k_l\times k_l\times 3\times 3}$ is the convolution weights of the Lth RandConv layer, which are sampled from a Gaussian distribution $\mathcal{N}(0, (1/3k_l^2))$, and the kernel size k_l is uniformly sampled from $\{1, 3, 5, 7\}$.

For style divergences, AdaIN [51] is a simple yet effective solution, which is widely used in the style transfer field and stylizes content images by aligning the style statistics of the content features with those of the style features. Given two inputs: the content information x and the style information z, AdaIN can stylize x via matching their channel-wise mean and variance, as (2). However, AdaIN requires specific style input as a reference, thus making it unable to handle varied style information in unseen target domains

$$\text{AdaIN}(x, z) = \sigma(z)\left(\frac{x - \mu(x)}{\sigma(x)}\right) + \mu(z). \tag{2}$$

Therefore, we devise a novel normalization layer, termed RandIN, for style randomization. RandIN only requires the content input and replaces the statistics from the style information [i.e., $\sigma(z)$ and $\mu(z)$ in (2)] in AdaIN with γ and β that are randomly initialized from $\mathcal{N}(0, \sigma_\gamma^2)$ and $\mathcal{N}(0, \sigma_\beta^2)$, respectively. The process of RandIN can be defined as

$$\text{RandIN}(x) = \gamma\left(\frac{x - \mu(x)}{\sigma(x)}\right) + \beta. \tag{3}$$

Besides, to prevent excessive semantic distortion of the generated data, TSRM also incorporates the residual connection [52] to mix with the source domain data in random proportions. And given the input x, the work mechanism of TSRM can be summarized as follows:

$$\text{TSRM}(x) = \alpha x + (1 - \alpha)\,\text{RandIN}(\text{Multi_RC}(x, L)) \tag{4}$$

where α is the mixing weight uniformly sampled from [0, 1).

In our SDG method, TSRM performs domain randomization of texture and style information to generate the auxiliary domain for each training epoch. Due to the fact that the key parameters of TSRM are randomized for each mini-batch during training, the auxiliary domain in one epoch not only differs from other epochs but also encompasses more than one data distribution. In other words, TSRM can offer a wide range of data distributions in a simple yet effective manner.

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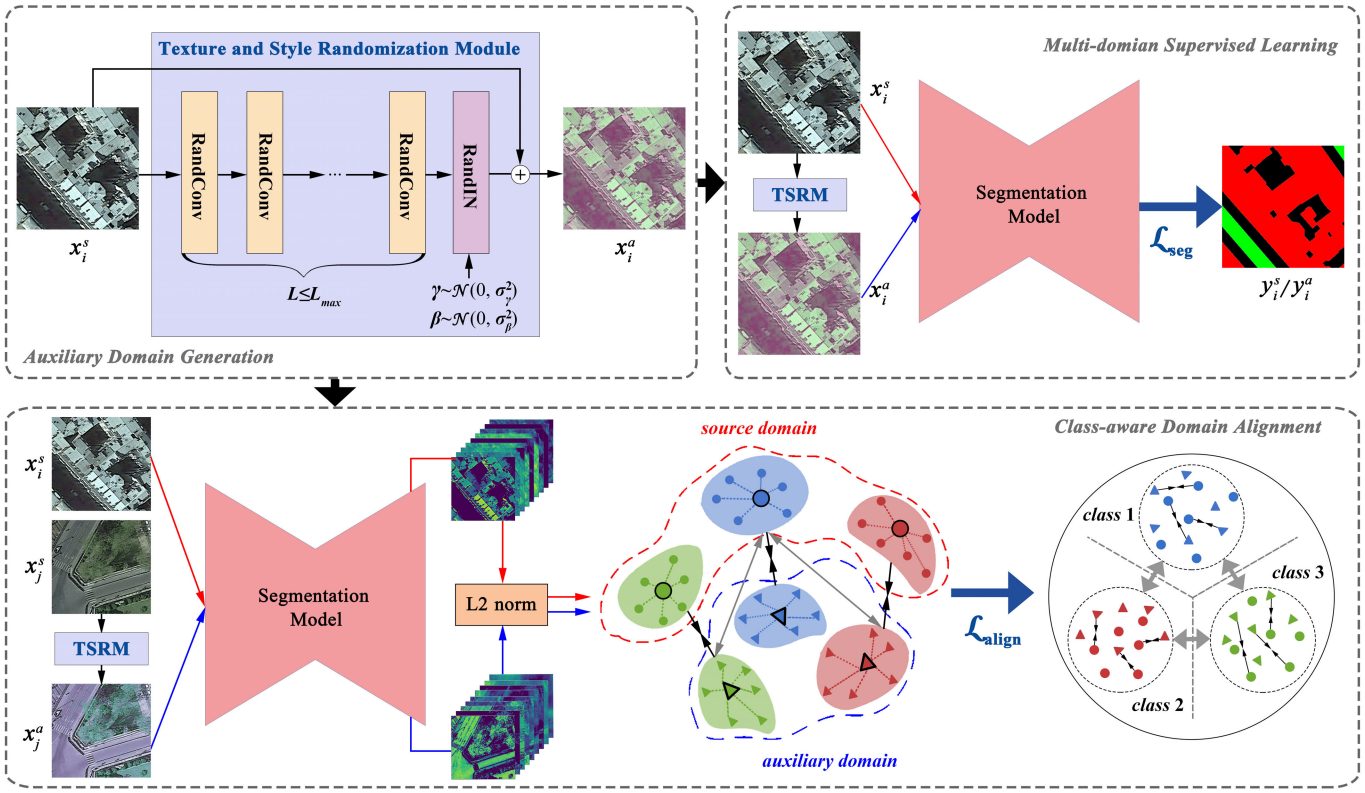


Fig. 2. CCDR framework. CCDR implements auxiliary domain generation with multiple RandConv layers and a novel normalization layer (RandIN), performs supervised learning with L_{seg} using the source domain and the generated auxiliary domain, and imposes L_{align} at the feature level for the class-aware regularization across domains.

C. Multidomain Supervised Learning

Our ultimate goal is to obtain a domain-agnostic segmentation model by exposing it to diverse domain variations. Multidomain supervised learning can empower the segmentation model with the ability to enhance interclass separability and intraclass similarity within each domain. Lots of existing SDG methods [25], [28], [29] utilize multidomain supervised learning to guide the segmentation model to focus more on domain-invariant representations. The cross-entropy loss is used as the cost function. During multidomain supervised learning, given the sample from the source or auxiliary domain (x, y), the segmentation loss can be defined as

$$\mathcal{L}_{\text{seg}}(x, y; \psi, \phi) = -\sum_{c=1}^{N_c} \delta(y = c) \log h_\phi\big(g_\psi(x)\big) \tag{5}$$

where $\delta(\cdot)$ is an indicator function that returns 1 if the argument is true, and 0 otherwise, and $h_\phi(\cdot)$ and $g_\psi(\cdot)$ are the classifier and the feature extractor of the segmentation model, respectively.

D. Class-Aware Domain Alignment

Due to the inherent nature of DNN-based models, which tend to rely on superficial features rather than global shape information for prediction, multidomain supervised learning is susceptible to over-fitting and inadequate to derive sufficient domain-agnostic representations. In addition, varying degrees of semantic distortion in the auxiliary domain generated by TSRM may be detrimental to class discriminative information that plays a vital role in remote sensing image segmentation tasks.

In order to circumvent the aforementioned limitations, improve the reliability of classification, and enhance the class discriminative ability of the segmentation model, it should not only focus on intraclass and interclass relationships within each domain but also appreciate semantic correlation across the domains. Therefore, in addition to clarifying classification boundaries within each domain via multidomain supervised learning, CCDR further inserts a class-aware regularization between the source and auxiliary domains.

Specifically, CCDR employs the triple loss [53] to implement the cross-domain constraint for class-aware domain alignment. Given the samples randomly sampled from the source and auxiliary domains, $(x_i^s, y_i^s) \sim \mathcal{D}^S$ and $(x_j^a, y_j^a) \sim \mathcal{D}^A$, the features before the classification layer are rescaled to the range $[0, 1]$ using the $L2$-norm layer, which allows obtaining the source domain representation $f_i^s \in \mathbb{R}^{H \times W \times N_f}$ and the auxiliary domain representation $f_j^a \in \mathbb{R}^{H \times W \times N_f}$.

After that, we calculate the class embeddings of the source domain sample $e_i^s = \{e_{i,c}^s\}_{c=1}^{N_c}$, where $e_{i,c}^s \in \mathbb{R}^{N_f}$ is the cth class embedding of (x_i^s, y_i^s), and can be formulated as (6). And the class embeddings of the auxiliary domain sample $e_j^a = \{e_{j,c}^a\}_{c=1}^{N_c}$ can be calculated in the same way

$$e_{i,c}^s = \frac{\sum_{c=1}^{N_c} \delta\big(y_i^s = c\big) \cdot f_i^s}{\sum_{c=1}^{N_c} \delta\big(y_i^s = c\big)}. \tag{6}$$

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Then, following the triplet loss, we pull together the embeddings that belong to the same class but different domains and push away the embeddings that belong to different classes and different domains. For instance, for the cth class, $e^s_{i,c}$, $e^a_{j,c}$, and $e^a_{j,c'}$ can be considered as the anchor, positive sample, and negative sample, respectively, where $c \neq c'$. And the cross-domain constraint in CCDR can be defined as

$$\mathcal{L}_{\text{align}}\left(x_i^s, x_j^a; \psi\right) = \frac{1}{N_c(N_c-1)} \sum_{c,c'} \delta\left(c \neq c'\right) \cdot \max\left\{0, d\left(e^s_{i,c}, e^a_{j,c}\right) - d\left(e^s_{i,c}, e^a_{j,c'}\right) + \xi\right\} \tag{7}$$

where $d(\cdot)$ is the L_2 distance function and ξ is a hyperparameter standing for the margin value.

E. Optimization

In order to facilitate the implementation of $\mathcal{L}_{\text{align}}$ and prevent the cheating issue caused by information leakage from batch normalization layers, the auxiliary domain samples within a min-batch are shuffled before training, and following the previous work [54], our method is trained with multiple GPUs. The detailed training procedure for CCDR is presented in Algorithm 1.

Algorithm 1 Learning Algorithm for CCDR

Input: Source dataset $\mathcal{D}^S = \left\{\left(x_i^s, y_i^s\right)\right\}_{i=1}^N$; hyperparameters $L_{\max}$, σ_γ, σ_β and ξ; learning rate η; batch size N_s.

Output: Well-trained remote sensing segmentation network, composed of the feature extractor $g_\psi(\cdot)$ and the classifier $h_\phi(\cdot)$.

Initialize network parameters ψ and ϕ.

while not converged **do**

- $\left\{\left(x_i^s, y_i^s\right)\right\}_{i=1}^{N_s} \sim \mathcal{D}^S$ // Sample a min-batch from the source domain
- **Texture and style randomization**:
- $L \sim \mathcal{U}(\{1, 2, \ldots, L_{\max}\})$ // The number of random convolution layers
- **for** l=1 **to** L **do**
  - $k_l \sim \mathcal{U}(\{1, 3, 5, 7\})$ // Kernel size
  - $\Theta_l \sim \mathcal{N}\left(0, \frac{1}{3k_l^2}\right)$ // Convolution weights
  - **for** i=1 **to** N_s **do**
    - $x^s_{i,l} = x^s_{i,l-1} * \Theta_l$ // Convolution operation
- $\gamma \sim \mathcal{N}\left(0, \sigma_\gamma^2\right)$, $\beta \sim \mathcal{N}\left(0, \sigma_\beta^2\right)$ // Style information
- $\alpha \sim \mathcal{U}[0, 1)$ // Mixing weight
- **for** i=1 **to** N_s **do**
  - $x_i^a = \alpha x_i^s + (1-\alpha)\left[\gamma\left(\frac{x^s_{i,L} - \mu(x^s_{i,L})}{\sigma(x^s_{i,L})}\right) + \beta\right]$
  - $y_i^a = y_i^s$
  - // Generate a min-batch from the auxiliary domain
- **Shuffling the min-batch from the auxiliary domain, and training the remote sensing image segmentation network**:
- $\mathcal{L}\left(x_i^s, y_i^s, x_i^a, y_i^a; \psi, \phi\right) = \frac{1}{2}\left[\mathcal{L}_{\text{seg}}\left(x_i^s, y_i^s; \psi, \phi\right) + \mathcal{L}_{\text{seg}}\left(x_i^a, y_i^a; \psi, \phi\right)\right] + \mathcal{L}_{\text{align}}\left(x_i^s, x_i^a; \psi\right)$
- $\mathcal{L}_{\text{all}} = \frac{1}{N_s}\sum_{i=1}^{N_s} \mathcal{L}\left(x_i^s, y_i^s, x_i^a, y_i^a; \psi, \phi\right)$
- $(\psi, \phi) \leftarrow (\psi, \phi) - \eta \nabla_{\psi,\phi} \mathcal{L}_{\text{all}}$ // Update network parameter

return $h_\phi \circ g_\psi$.

IV. Experimental Study

A. Dataset Description

1) WHU Dataset: This dataset [55] provides a large amount of pixel-level annotations for *building* extraction, and there are three subdatasets as follows.

The Aerial dataset contains 8189 aerial images collected from New Zealand with a size of 512 × 512 pixels and a spatial resolution of 0.3 m, which distributes various and versatile architecture types of buildings with different colors, sizes, and usages.

The Satellite I dataset is collected from cities around the world and from various remote sensing resources, including QuickBird, Worldview series, IKONOS, and ZY-3, and contains 204 images with a size of 512 × 512 pixels, and the spatial resolutions vary from 0.3 to 2.5 m.

The Satellite II dataset is constructed by six neighboring satellite images covering 550 m^2 in East Asia with 2.7 m spatial resolution. And there are 17 388 pixel-level annotated patches with a size of 512 × 512 pixels.

In the following, we evaluate our method on the WHU dataset, where the Aerial dataset (A) is employed as the source domain, and the Satellite I (S_1) and Satellite II (S_2) datasets are utilized as the target domains. During training, we adopt the cross-validation strategy, and each subdataset is randomly split into training and test sets at a ratio of 3:2.

2) OM Dataset: This dataset [56] collects aerial images from six cities and provides pixel-level annotation for two categories: *building* and *road*. There are six subdatasets according to the geographic locations, including the Berlin, Chicago, Paris, Postdam, Tokyo, and Zurich datasets. The Postdam dataset is available from the ISPRS semantic labeling benchmark [57] and contains 24 images with a size of 3296 × 3296 pixels. Other datasets are collected from Google Maps, and their annotations are automatically generated from OpenStreetMap [58] (OSM, http://www.openstreetmap.org), and the more details are as shown in Table I.

In experiments on the OM dataset, we adopt the Paris (Pa) dataset as the source domain and evaluate our SDG method on the target domains, including the Berlin (B), Chicago (C), Postdam (Po), Tokyo (T), and Zurich (Z) datasets. The partitioning of the training and test sets on the OM dataset is consistent with that of the WHU dataset.

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TABLE I
DETAILS OF EACH SUBDATASET IN THE OM DATASET

Dataset Name	Size ($number \times height \times width$)
Berlin	200 × 2611 × 2453
Chicago	458 × 3326 × 2560
Paris	625 × 3328 × 3072
Postdam	24 × 3296 × 3296
Tokyo	1 × 2500 × 2500
Zurich	364 × 3328 × 2816

TABLE II
TRAINING PARAMETERS OF CCDR

Dataset	$L_{\max}$	σ_γ	σ_β	ξ	N_s
WHU	10	1	1	0.5	6
OM	10	1	1	0.5	4

B. Implementation

1) Training Detail: We implement our method with the PyTorch framework on four 24GB GPUs (NVIDIA GeForce RTX 3090). In experiments on the WHU dataset, the Adam optimizer is used to train the model with an initial learning rate of 1e-4, betas of 0.9 and 0.999, and weight decay of 1e-4. In experiments on the OM dataset, the training samples on each subdataset are randomly cropped to a size of 512 × 512 pixels, the stochastic gradient descent (SGD) with an initial learning rate of 1e-2, the momentum of 0.9, and weight decay of 1e-3 are used as the optimizer. Other training parameters mentioned in Algorithm 1 are set as Table II.

2) Accuracy Evaluation Criteria: F_1 is used for accuracy evaluation, as (10)

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{10}$$

where precision and recall can be calculated as (8) and (9). TP and TN are the number of positive and negative samples correctly predicted, respectively; FP and FN are the number of negative and positive samples incorrectly predicted, respectively.

3) Complexity Evaluation Criteria: FLOPs and Params are adopted to evaluate the algorithm complexity. FLOPs is the number of floating-point operations, and Params is the number of trainable parameters. In the following, the complexity evaluation criteria are calculated under the condition that the input size is 512 × 512 × 3, the batch size is 1, and the number of categories is 2.

4) Auxiliary Domain Evaluation Criteria: Since real target domains are unaccessible in the SDG paradigm, the well-studied full-reference image quality assessment (IQA) criteria [59], [60], [61] are not applicable for the generated auxiliary domain evaluation. Therefore, blind/referenceless image spatial quality evaluator (BRISQUE) [62] and style statistics, both not requiring reference images, are employed for auxiliary domain evaluation from two perspectives of texture and style, and two dimensions of magnitude and diversity.

The BRISQUE score can reflect spatial distortion based on natural scene statistics in a reference-less manner. Therefore, we calculate the BRISQUE scores of the source and auxiliary domains to evaluate the texture distortions of the auxiliary domain. Their differences can stand for the texture distortion magnitude, and the standard deviation of their differences can reflect the texture distortion diversity.

Specifically, given the source domain images $\{x_i^s\}_{i=1}^N$ and the auxiliary domain images $\{x_i^a\}_{i=1}^N$, the differences in BRISQUE scores $\{\Delta_{\text{texture}}^i\}_{i=1}^N$ can be calculated as

$$\Delta_{\text{texture}}^i = \left\|\text{BRISQUE}\left(x_i^a\right) - \text{BRISQUE}\left(x_i^s\right)\right\|_1 \tag{11}$$

where $\|\cdot\|_1$ is the L_1 distance function and BRISQUE$(\cdot)$ is the pretrained prediction model provided by Mittal et al. [62] that is available online: http://live.ece.utexas.edu/research/quality/BRISQUE_release.zip. Then, the texture distortion magnitude criterion M_{texture}, and the texture distortion diversity criterion D_{texture} can be formulated as follows:

$$M_{\text{texture}} = \frac{1}{N}\sum_{i=1}^{N}\Delta_{\text{texture}}^i \tag{12}$$

$$D_{\text{texture}} = \sqrt{\frac{\sum_{i=1}^{N}\left(\Delta_{\text{texture}}^i - M_{\text{texture}}\right)^2}{N}}. \tag{13}$$

However, M_{texture} and D_{texture} cannot reflect the style distortions of the auxiliary domain. Therefore, we collect the style statistics of the source and auxiliary domains, i.e., $\{u_i^s\}_{i=1}^N$ and $\{u_i^a\}_{i=1}^N$, respectively. u_i^a consists of the mean and standard deviation of each channel, namely, $u_i^a = [\mu_i^a, \sigma_i^a]$, where $u_i^a \in \mathbb{R}^{2\times 3}$, $\mu_i^a \in \mathbb{R}^{1\times 3}$, and $\sigma_i^a \in \mathbb{R}^{1\times 3}$. And u_i^s can be obtained in the same way.

Akin to the texture distortion evaluation, we calculate the differences in style statistics between the source and auxiliary domains, as

$$\Delta_{\text{style}}^i = \|u_i^s - u_i^a\|_1. \tag{14}$$

Then, the style distortion magnitude criterion M_{style}, and the style distortion diversity criterion D_{style} can be formulated as

$$M_{\text{style}} = \frac{1}{N}\sum_{i=1}^{N}\Delta_{\text{style}}^i \tag{15}$$

$$D_{\text{style}} = \sqrt{\frac{\sum_{i=1}^{N}\left(\Delta_{\text{style}}^i - M_{\text{style}}\right)^2}{N}}. \tag{16}$$

C. Comparison Study

In comparative experiments, UNet [63] is employed as the baseline network, and the following state-of-the-art SDG methods are used for comparison.

1) **Empirical risk minimization (ERM)** [64] is the baseline method trained only with the cross-entropy loss and the labeled data from the source domain.

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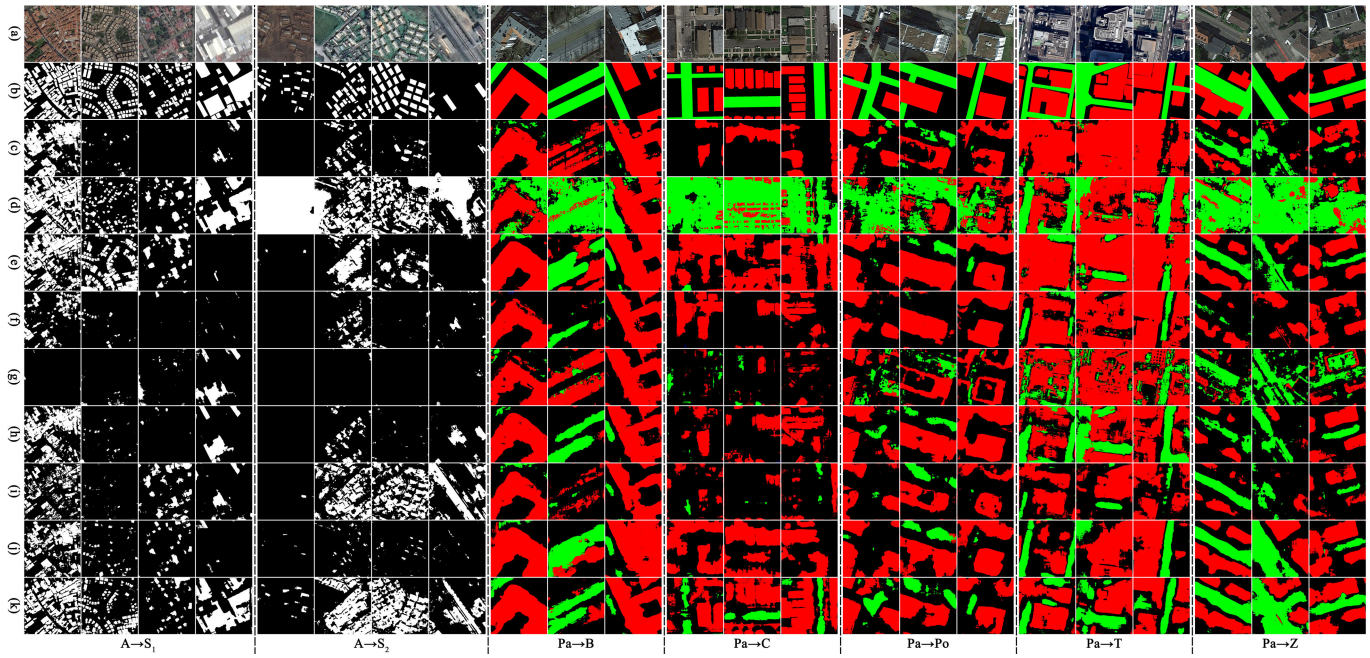


Fig. 3. Segmentation results of different SDG methods on the WHU and OM datasets, where black means *background* on the two datasets, white stands for *building* on the WHU dataset, and red and green represent *building* and *road* on the OM dataset, respectively. (a) Image. (b) Ground truth. (c) ERM. (d) AugMix. (e) RCKL. (f) MixStyle. (g) ADA. (h) M-ADA. (i) L2D. (j) PDEN. (k) CCDR.

2) **AugMix** [31] mixes multiple augmented images from different augmentation functions, and minimizes the JS divergence among the predictions of the original sample and its augmented versions.
3) **RandConv with KL divergence (RCKL)** [33] expands the coverage of the source domain with a single RandConv layer and narrows the KL divergence between the predictions of various domains and their average.
4) **MixStyle** [32] argues that there are several subdomains within one source domain, and swaps style statistics of features from different samples to improve the generalization.
5) **ADA** [25] minimizes the distance of semantic space and maximizes the classification loss to generate adversarial samples for generalization training.
6) **M-ADA** [26] incorporates WAE into ADA to generate more diverse auxiliary domains, and embraces meta-learning to further enhance model generalization.
7) **Learning to diversify (L2D)** [30] deploys five style-complement modules, and enhances the generalization power of the task model in an adversarial min-max manner based on mutual information.
8) **PDEN** [28] progressively learns 20 generators for safe and effective data generation, and gathers domain-invariant feature representation via contrastive learning.

The hardware environment of all aforementioned methods is consistent with CCDR, and their hyperparameters follow their original papers. Fig. 3 sketches their segmentation results on the WHU and OM datasets. Table III shows their recognition accuracy results, where the optimal values are highlighted in bold, and the suboptimal values of all methods except ERM are underlined. Table IV compares their algorithm complexity via $FLOPs$ and $Params$. The detailed analysis is presented as follows.

First, we intuitively compare these SDG methods through their segmentation results. As shown in Fig. 3(c), the segmentation results of ERM are far from satisfactory, influenced by the complex domain shift in remote sensing. On the WHU dataset, CCDR [Fig. 3(k)] obviously outperforms other SDG methods in both A→S_1 and A→S_2 experiments, AugMix [Fig. 3(d)] performs high false detections, while other state-of-the-art SDG methods [Fig. 3(e)–(j)] exhibit high missed detections. On the OM dataset, AugMix misidentifies a lot of *background* as *road*, as shown in Fig. 3(d). Although RCKL and MixStyle avoid a wide range of false detections, as shown in Fig. 3(e) and (f), their missed detections on the *road* and false detections on the *building* are notably more pronounced compared to CCDR. ADA exhibits severely missed detections, as shown in Fig. 3(g), and cannot effectively distinguish *road* with *building*. M-ADA has great improvement over ADA, as shown in Fig. 3(h), but is still inferior to CCDR. And compared to CCDR, PDEN, and L2D cannot effectively extract *road*, as shown in Fig. 3(i) and (j), and L2D also has high missed detections on the *building*. In comparison, the segmentation results of CCDR are most similar to the ground truths, as shown in Fig. 3(k).

Then, the generalization performance of CCDR is quantitatively evaluated via the comparison of accuracy results, as shown in Table III. Specifically, unlike other SDG methods that suffer from varying degrees of performance degradation compared to ERM, CCDR can improve the F_1 of *building* over 20.0% in the A→S_1, A→S_2, Pa→C, and Pa→Z experiments, and increase the F_1 of *road* over 13.0% in the Pa→C, Pa→Po, and Pa→Z experiments. And CCDR is slightly inferior to ERM only when extracting *building* in the Pa→Po experiment.

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TABLE III
ACCURACY COMPARISON OF DIFFERENT SDG METHODS ON THE WHU AND OM DATASETS

Method	A→S$_1$	A→S$_2$	Pa→B		Pa→C		Pa→Po		Pa→T		Pa→Z	
	Building	*Building*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*
ERM [64]	0.2780	0.2390	0.6271	0.3206	0.4100	0.0943	0.6048	0.2360	0.7684	0.1964	0.4159	0.2893
AugMix [31]	0.4387	0.1363	0.5231	0.3683	0.1112	0.2793	0.2432	0.3046	0.6520	0.5796	0.1302	0.3304
RCKL [33]	0.4298	0.1728	0.6409	0.3494	0.6110	0.1656	0.6340	0.2046	0.7659	0.3449	0.5781	0.3086
MixStyle [32]	0.2775	0.0735	0.6431	0.1926	0.3775	0.0020	0.6014	0.0901	0.7479	0.2110	0.4700	0.0997
ADA [25]	0.2580	0.0215	0.5775	0.2265	0.2682	0.0072	0.5293	0.2551	0.6820	0.4138	0.2129	0.3161
M-ADA [26]	0.3150	0.1117	0.6554	0.3950	0.3987	0.0376	0.5563	0.2371	0.6192	**0.6334**	0.4490	0.2848
L2D [30]	0.4410	0.2223	0.5999	0.2289	0.3247	0.1231	0.5309	0.2029	0.7622	0.0932	0.3798	0.2827
PDEN [28]	0.4013	0.1516	**0.6826**	0.3624	0.5664	0.0651	**0.6401**	0.2717	0.6198	0.4512	**0.6637**	0.3269
CCDR	**0.4876**	**0.4598**	0.6345	**0.4023**	**0.6247**	**0.3146**	0.6017	**0.3696**	**0.7762**	0.4053	0.6217	**0.4452**

TABLE IV
COMPLEXITY COMPARISON OF DIFFERENT SDG METHODS

	ERM [64]	AugMix [31]	RCKL [33]	MixStyle [32]	ADA [25]	M-ADA [26]	L2D [30]	PDEN [28]	CCDR
Params (MB)	39.2380	39.2380	39.2380	39.2380	40.0244	198.1043	39.2388	39.7608	39.2380
FLOPs (GB)	267.3156	801.9486	801.9467	534.6312	1336.5818	1344.4147	1071.8944	2126.3614	534.8836

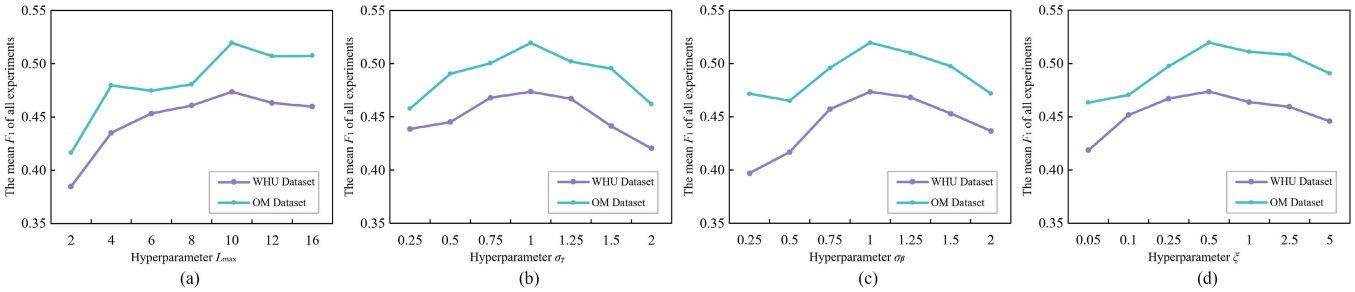


Fig. 4. Sensitivity analysis of hyperparameters in CCDR. (a) $L_{\max}$. (b) σ_γ. (c) σ_β. (d) ξ.

After that, we further compare CCDR with other SDG methods. Obviously, CCDR obtains optimal performance in most experiments and has impressive advantages over other SDG methods. For instance, compared to the best values among other SDG methods, CCDR can improve the F_1 of *building* by more than 20.0% in the A→S2 experiment, and increase the F_1 of *road* by over 10.0% in the Pa→Z experiment. Overall, the noticeable improvement of CCDR over ERM and other SDG methods demonstrates the effectiveness and advantages of our method in dealing with the complex domain shift in remote sensing images.

Afterward, we utilize FLOPs and Params to analyze the algorithm complexity of our method, as shown in Table IV. To be concrete, CCDR is in line with ERM in terms of Params, and its FLOPs are twice as many as those of ERM. Compared with other SDG methods, the FLOPs and Params of CCDR are comparable with those of MixStyle, which does not require synthesizing additional auxiliary domains. Although AugMix and RCKL are on par with ERM and CCDR in terms of Params, as they perform simple auxiliary domain generation without additional training, their FLOPs are considerably higher than those of CCDR. Furthermore, compared to ADA, M-ADA, L2D, and PDEN, CCDR has noticeable advantages in both Params and FLOPs. ADA and M-ADA both adopt ADA with complicated backward propagation, and M-ADA further introduces the meta-learning strategy and WAE model for data generation, which results in their obvious drawbacks in Params and FLOPs. L2D and PDEN implement several lightweight generators for auxiliary domain generation, which do not sharply increment their Params, but their intricate training pipelines bring substantial increases in FLOPs.

Altogether, in most experiments, CCDR shows significant improvement over ERM and surpasses other state-of-the-art SDG methods without the cumbersome data generation process and training pipeline, which illustrates the superiority of CCDR in remote sensing image segmentation tasks.

D. Sensitivity Analysis

This section conducts sensitivity analyses on the hyperparameters in CCDR, which have been mentioned in Algorithm 1, through the mean F_1 of all experiments on each dataset, as depicted in Fig. 4.

1) Hyperparameter of Multi_RC: We implement different numbers of RandConv layers with uniform sampling from 1 to $L_{\max}$ for each mini-batch, as described in Algorithm 1. Fig. 4(a) shows that the performance of CCDR has been substantially improved with the increases of $L_{\max}$, under the condition of $L_{\max} \leq 10$. However, $L_{\max}$ and F_1 do not always present a positive correlation, as too many RanConv layers may impair semantics. When $L_{\max}$ is over 10, there are varying degrees of performance degradation on each dataset.

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TABLE V
ABLATION STUDY OF CCDR ON THE WHU AND OM DATASETS

Method	Variants	Multi_RC	RandIN	$\mathcal{L}_{\text{align}}$	A→S$_1$	A→S$_2$	Pa→B		Pa→C		Pa→Po		Pa→T		Pa→Z	
					Building	*Building*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*
ERM	-				0.2780	0.2390	0.6271	0.3206	0.4100	0.0943	0.6048	0.2360	0.7684	0.1964	0.4159	0.2893
CCDR	w/ Multi_RC	✓			0.4318	0.2036	0.6318	0.2325	0.5143	0.2276	0.5869	0.1490	0.7320	0.2177	0.4672	0.3364
	w/ RandIN		✓		0.4460	0.3037	0.5801	0.1741	0.5765	0.1674	0.5413	0.0925	0.7602	0.1135	0.4736	0.1888
	w/o $\mathcal{L}_{\text{align}}$	✓	✓		0.4099	0.3351	0.6192	0.2547	0.5673	0.2231	0.5737	0.1893	0.7661	0.1938	0.5271	0.2658
	w/o RandIN	✓		✓	0.4378	0.3487	0.6366	0.3588	0.5199	0.3185	0.6050	0.2364	0.7827	0.3514	0.5713	0.4339
	w/o Multi_RC		✓	✓	0.4864	0.3455	0.6537	0.3075	0.5825	0.2279	0.5596	0.3399	0.7767	0.1303	0.5123	0.4368
	-	✓	✓	✓	0.4876	0.4598	0.6345	0.4023	0.6247	0.3146	0.6017	0.3696	0.7762	0.4053	0.6217	0.4452

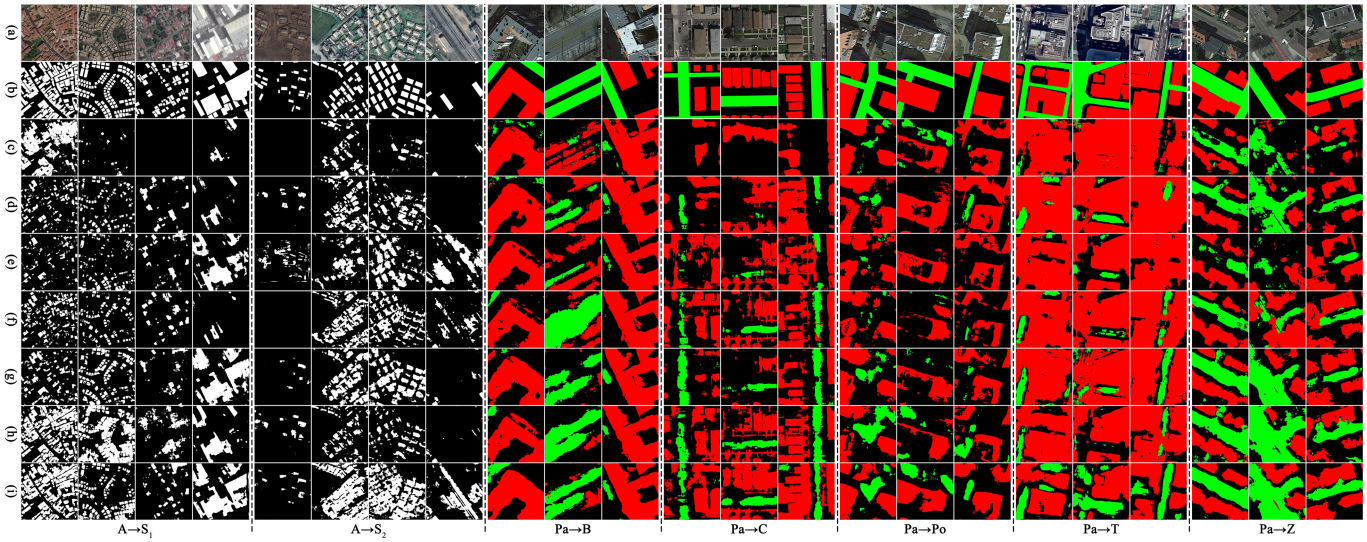


Fig. 5. Segmentation results of CCDR in the ablation study on the WHU and OM datasets. (a) Image. (b) Ground truth. (c) ERM. (d) CCDR w/ Multi_RC. (e) CCDR w/ RandIN. (f) CCDR w/o $\mathcal{L}_{\text{align}}$. (g) CCDR w/o RandIN. (h) CCDR w/o Multi_RC. (i) CCDR.

2) Hyperparameters of RandIN: σ_γ and σ_β are used to sample the style information of RandIN, namely, $\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$ and $\beta \sim \mathcal{N}(0, \sigma_\beta^2)$. Within a certain range, the larger the values of σ_γ and σ_β, the higher the diversity of the generated auxiliary domain, the greater the improvement of model generalization. However, as a special type of affine transformation, RandIN may bring about false distortion or saturation when σ_γ and σ_β are larger than 1, resulting in performance degradation, as shown in Fig. 4(b) and (c).

3) Hyperparameter of $\mathcal{L}_{align}$*:* $\mathcal{L}_{\text{align}}$ encourages that dissimilar pairs be distant from any similar pairs by at least a certain margin value ξ. Whereas if ξ is too large, it will lead to nonconvergence. If ξ is too small, it will make it challenging to distinguish between positive and negative samples. And we chose ξ to be 0.5, as it shows the highest performance on both WHU and OM datasets, as shown in Fig. 4(d).

E. Ablation Study

To examine the contribution of different components within CCDR, extensive ablation experiments are conducted on the WHU and OM datasets using UNet as the baseline network. The following variants of CCDR are implemented for ablation analysis.

1) **CCDR w/ Multi_RC** only uses Multi_RC for texture randomization in auxiliary domain generation and only performs multidomain supervised learning.
2) **CCDR w/ RandIN** only implements style randomization with RandIN during data generation and trains the segmentation model using the cross-entropy loss and labeled data from the source and auxiliary domains, without employing domain alignment.
3) **CCDR w/o $\mathcal{L}_{\text{align}}$** applies multidomain learning to the segmentation model using the source domain and the auxiliary domain generated by TSRM.
4) **CCDR w/o RandIN** performs multidomain learning and class-aware domain alignment, where the adopted auxiliary domain is generated by Multi_RC.
5) **CCDR w/o Multi_RC** removes Multi_RC from TSRM, which is the only thing different from CCDR.

Table V displays the accuracy results of ERM, CCDR, and its variants, and Fig. 5 illustrates their segmentation results on the WHU and OM datasets.

First, we focus on Variant-1 (CCDR w/ Multi_RC) and Variant-2 (CCDR w/ RandIN), in order to discuss the special roles of Multi_RC and RandIN within TSRM. Especially, compared to other SDG methods, as shown in Tables III and V, these variants can realize comparable or even superior performance. The performance of Variant-1 and Variant-2 varies with the main factor causing domain shifts in different experiments. Besides, these variants without class-aware domain alignment do not consistently outperform ERM in all cases, and decreases in F_1 are very severe in some experiments (even over 10.0%). Altogether, these experimental

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TABLE VI
SCALABILITY STUDY OF CCDR ON THE WHU AND OM DATASETS

Method	Baseline	A→S$_1$	A→S$_2$	Pa→B		Pa→C		Pa→Po		Pa→T		Pa→Z	
		Building	*Building*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*	*Building*	*Road*
ERM	UNet [63]	0.2780	0.2390	0.6271	0.3206	0.4100	0.0943	0.6048	0.2360	0.7684	0.1964	0.4159	0.2893
	DeepLabv3+ [65]	0.0631	0.0901	0.5819	0.3459	0.2537	0.0875	0.4498	0.2389	0.5192	0.3594	0.5726	0.2481
	SCAttNet [1]	0.1120	0.0973	0.5810	0.3273	0.2859	0.0285	0.4382	0.2460	0.7342	0.2543	0.4978	0.1955
CCDR	UNet [63]	0.4876	0.4598	0.6345	0.4023	0.6247	0.3146	0.6017	0.3696	0.7762	0.4053	0.6217	0.4452
	DeepLabv3+ [65]	0.2719	0.2350	0.5360	0.3674	0.3295	0.4600	0.5593	0.2481	0.6158	0.3813	0.5346	0.3258
	SCAttNet [1]	0.3776	0.2298	0.6362	0.4152	0.3982	0.2028	0.4901	0.2889	0.7547	0.4381	0.4899	0.4069

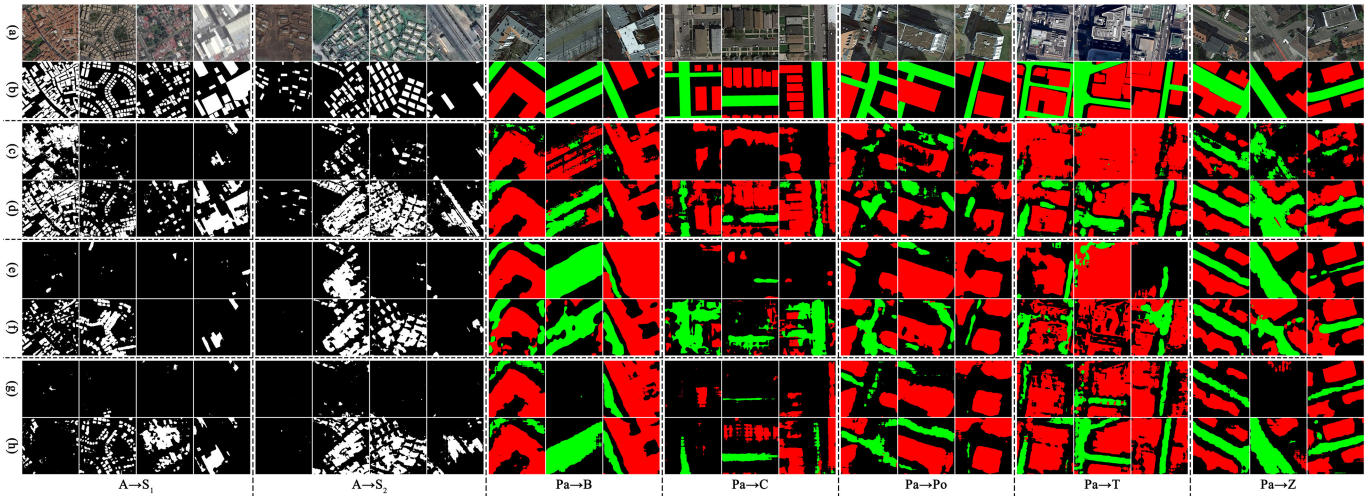


Fig. 6. Segmentation results of CCDR in the scalability study on the WHU and OM datasets. (a) Image. (b) Ground truth. (c) Results of ERM using UNet as the baseline. (d) Results of CCDR using UNet as the baseline. (e) Results of ERM using DeepLabv3+ as the baseline. (f) Results of CCDR using DeepLabv3+ as the baseline. (g) Results of ERM using SCAttNet as the baseline. (h) Results of CCDR using SCAttNet as the baseline.

results demonstrate that Multi_RC and RandIN within TSRM have significant advantages in simulating domain shift in remote sensing, but class discriminative information may be damaged in this completely random data generation process, leading to performance degradation compared to ERM in some cases.

Subsequently, we further explore the effect of $\mathcal{L}_{\text{align}}$ by comparing Variant-3 (CCDR w/o $\mathcal{L}_{\text{align}}$) with CCDR, Variant-1 with Variant-4 (CCDR w/o RandIN), and Variant-2 with Variant-5 (CCDR w/o Multi_RC). Experimental results illustrate that after applying $\mathcal{L}_{\text{align}}$ for class-aware domain alignment, the performance is improved to varying degrees in all experiments, which confirms the significance of performing class-aware domain alignment, and the effectiveness of our cross-domain constraint.

Finally, we discuss the necessity of implementing TSRM rather than only Multi_RC or RandIN. Theoretically, TSRM more comprehensively takes into account the two factors causing domain shift in remote sensing, i.e., texture and style divergences. Empirically, although Variant-3 does not possess an absolute advantage over Variant-1 and Variant-2, CCDR achieves superior performance over Variant-4 and Variant-5 after imposing the cross-domain constraint. Therefore, we contend that TSRM can facilitate more diverse data generation than Multi_RC or RandIN. Although TSRM may have a negative impact on the extraction of class discriminative information, class-aware domain alignment via $\mathcal{L}_{\text{align}}$ can effectively reduce the generated adverse effects and unleash its advantages.

F. Scalability Study

To investigate the applicability of CCDR on different baseline networks, two classical segmentation models in the computer vision field, namely, UNet [63] and DeepLabv3+ [65], as well as a remote sensing image segmentation model, termed SCAttNet [1], are employed for the scalability study.

Table VI presents the accuracy results of the three segmentation models before and after using CCDR, and Fig. 6 clarifies their segmentation results on the WHU and OM datasets. The detailed analysis is as follows.

On the WHU dataset, CCDR can substantially improve the accuracy of each segmentation model, and the increases reach 13.0%–22.1%.

On the OM dataset, for each segmentation model, CCDR has an obvious advantage in *road* extraction over ERM in all experiments. Besides, for *building* extraction on the OM dataset, although CCDR does not consistently outperform ERM when using each segmentation model, the improvement of CCDR over ERM is still remarkable in most experiments. For example, for the F_1 of *building*, CCDR can improve UNet by above 20.0% in the Pa→C and Pa→Z experiments,

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TABLE VII
QUALITATIVE EVALUATION OF AUXILIARY DOMAINS IN DIFFERENT SDG METHODS

	AugMix [31]	RCKL [33]	ADA [25]	M-ADA [26]	L2D [30]	PDEN [28]	CCDR
M_{texture}	4.8041	6.3720	4.3509	9.2984	9.9173	21.9535	14.1687
D_{texture}	3.4161	6.0020	3.2044	8.9368	9.4079	9.2395	18.4665
M_{style}	13.7462	18.9699	0.8142	15.3226	19.2181	35.4628	46.4043
D_{style}	8.5801	9.3320	0.3464	5.0465	5.4457	2.9189	9.8698

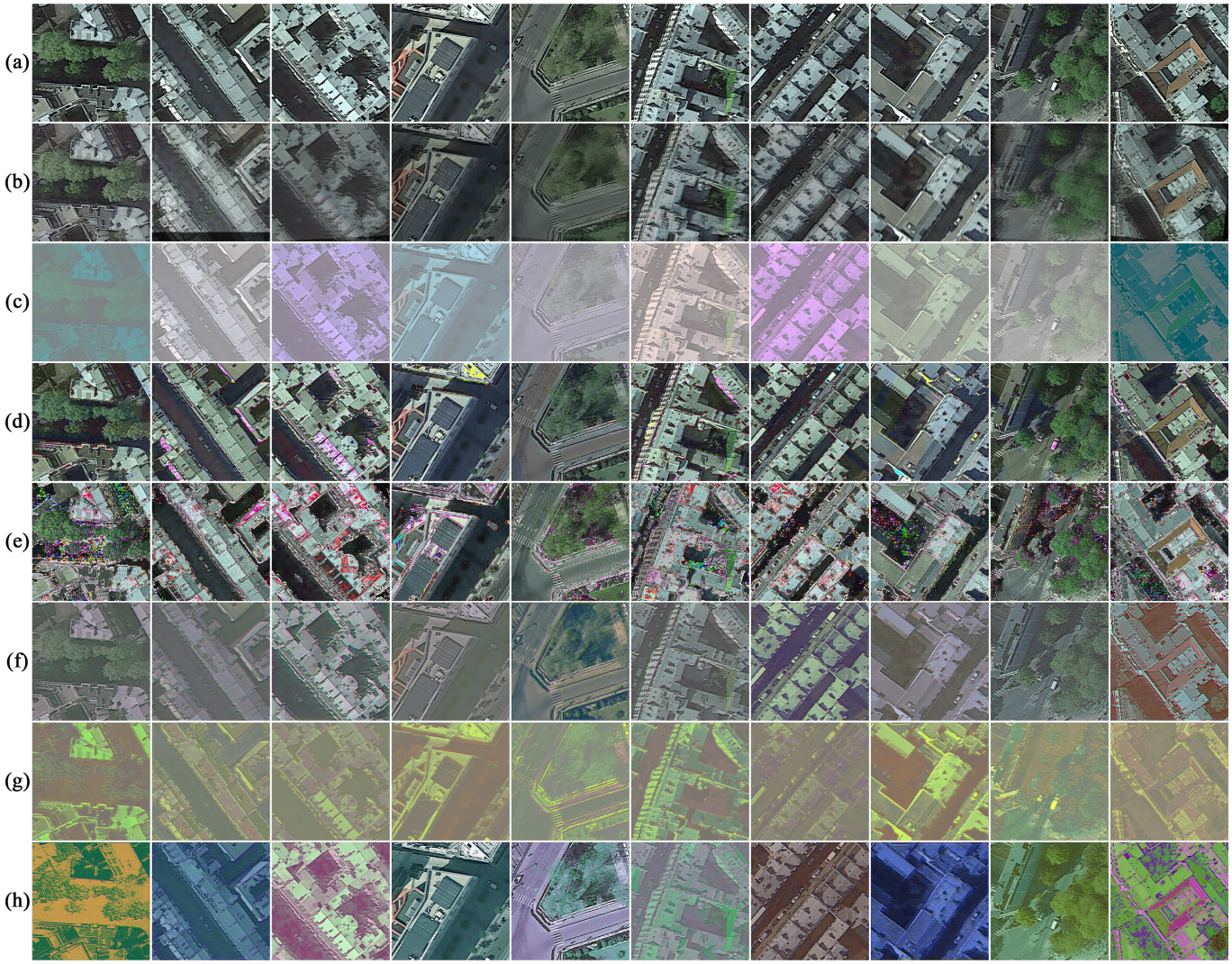


Fig. 7. Visualizations of generated data from different SDG methods on the OM dataset. (a) Images from the source domain. (b) AugMix. (c) RCKL. (d) ADA. (e) M-ADA. (f) L2D. (g) PDEN. (h) CCDR.

DeepLabv3+ by about 10.0% in the Pa→Po and Pa→T experiments, and SCAttNet by above 5.0% in the Pa→B, Pa→C, and Pa→Po experiments.

G. Visualization Study

To more intuitively expose the proposed SDG method, the visualization study is conducted from the following two perspectives.

1) Data Visualization: The auxiliary domain simulates domain shift through its distortions compared to the source domain. Therefore, the greater the diversity and magnitude of distortions (whether texture or style) without severely compromising semantics, the more helpful it is for the task model to learn domain-invariant representations. This section displays the auxiliary domains generated from different SDG methods on the OM dataset, as shown in Fig. 7, and presents the performance of these auxiliary domains on four evaluation criteria, as shown in Table VII.

Especially, AugMix adopts a random combination of traditional data augmentation functions to generate the auxiliary domain, which results in its limited texture and style distortions, as shown in Fig. 7(b), and its four criteria are only better than those of ADA.

The synthesized data from RCKL [Fig. 7(c)] has competitive performance on the diversity and magnitude of style

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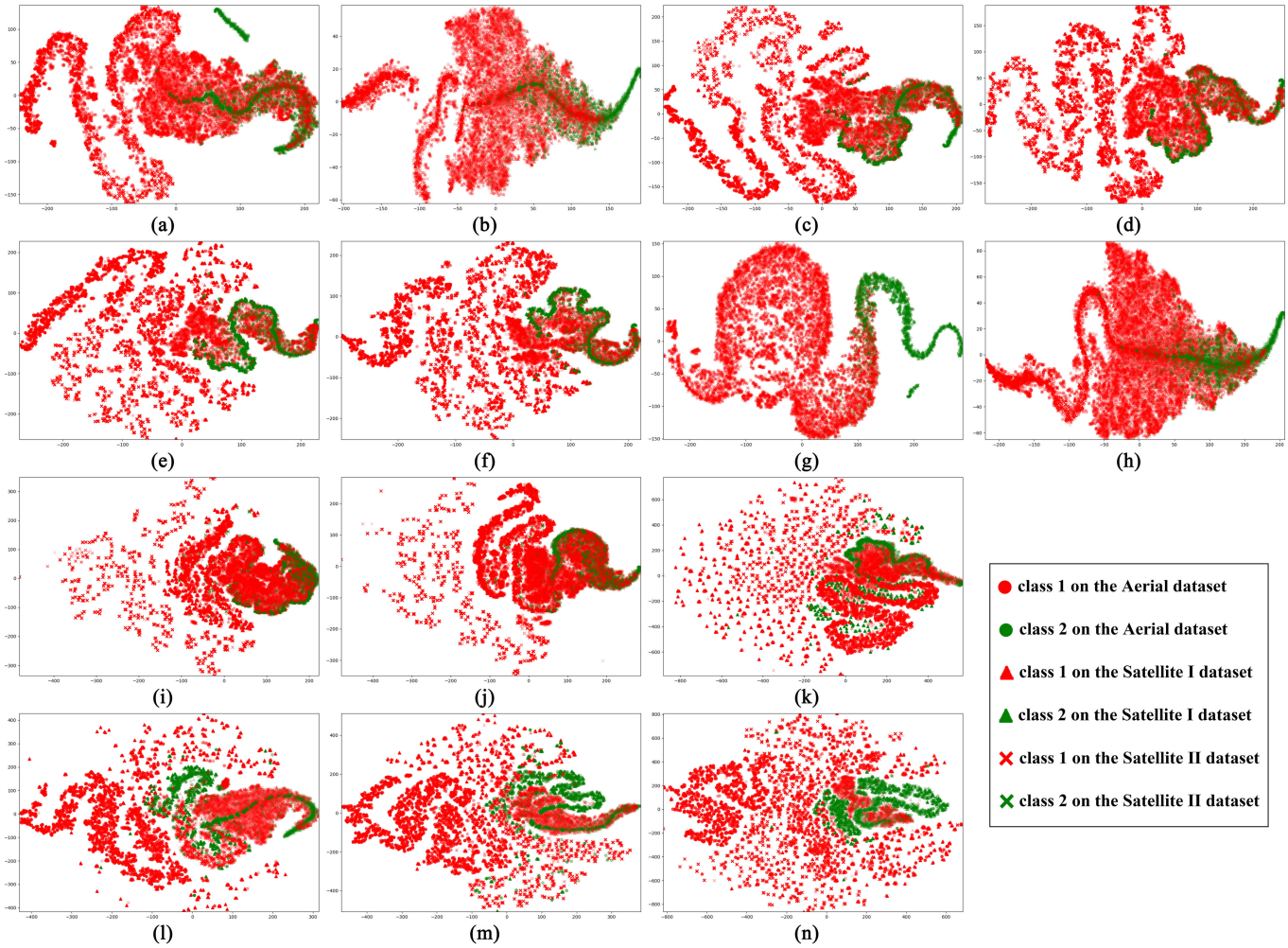


Fig. 8. t-SNE visualizations of the features extracted by different SDG methods on the WHU dataset, where class 1 means *background*, and class 2 is *building*. (a) ERM. (b) AugMix. (c) RCKL. (d) MixStyle. (e) ADA. (f) M-ADA. (g) L2D. (h) PDEN. (i) CCDR w/ Multi_RC. (j) CCDR w/ RandIN. (k) CCDR w/o $\mathcal{L}_{\text{align}}$. (l) CCDR w/o RandIN. (m) CCDR w/o Multi_RC. (n) CCDR.

distortions, its D_{style} is second only to that of CCDR, and its M_{style} is comparable with that of L2D (Table VII). Besides, although RCKL adopts RandConv, known for distorting local texture, it is inferior to M-ADA in terms of M_{texture} and D_{texture} since a single RanConv layer cannot yield adequate texture distortions.

Inspired by adversarial attacks, ADA only appends worst case perturbations for auxiliary domain generation, as shown in Fig. 7(d), and performs the worst on four evaluation criteria, as shown in Table VII. Owing to the WAE and meta-learning, the auxiliary domain from M-ADA has significant improvement compared to ADA, as shown in Fig. 7(e). However, M-ADA is still inferior to AugMix and RCKL in terms of style distortions, whether M_{style} or D_{style} as shown in Table VII.

Benefiting from several trainable data generation modules, the auxiliary domain from L2D is competitive in texture and style distortions, as shown in Fig. 7(f). Although L2D is still worse than AugMix and RCKL in terms of the style distortion diversity (D_{style}), L2D obtains suboptimal D_{texture}, and its M_{texture} and M_{style} are also second only to those of CCDR and PDEN, as shown in Table VII.

Fig. 7(g) illustrates the generated images from PDEN, which have evident advantages in texture distortions, but lack style diversify, despite the implementation of multiple generators. And the quantitative results in Table VII also indicate that PDEN has significant performance on the distortion magnitude of style and texture, but its D_{style} is only better than that of ADA.

And evidently, as shown in Fig. 7(h), the generated data from CCDR transcends that from RCKL in terms of distortion diversity and is comparable to that from PDEN in terms of distortion magnitude. The D_{texture}, M_{style}, and D_{style} of CCDR reach the peak, as shown in Table VII, and its M_{texture} is second only to that of PDEN, which means that the generated auxiliary domain from CCDR exhibits noticeable performance on the magnitude and diversity of texture and style distortions.

In summary, data visualization results demonstrate that the domain randomization module in CCDR, namely, TSRM, has remarkable advantages in auxiliary domain generation.

2) Feature Visualization: The goal of SDG is to tackle performance degradation due to data distribution differences. And whether the distribution of features extracted by the SDG method is influenced by domain information can intuitively provide feedback on their effectiveness. Therefore, we focus on the feature before the classification layer and visualize features extracted by different SDG methods with the t-distributed

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stochastic neighbor embedding (t-SNE) method, which can visualize high-dimensional data by giving each datapoint a location in a 2-D or 3-D map. The feature visualization is performed on the WHU dataset, as shown in Fig. 8.

In ERM, as shown in Fig. 8(a), the features from the same class but different domains typically exhibit distinct distributions within each target domain (Satellite I and Satellite II) and across domains, and the features from different classes are easily diffuse in each other. After using AugMix, RCKL, MixStyle, ADA, M-ADA, and PDEN [Fig. 8(b)–(f) and 8(h)] to weaken the interference of domain information on cross-domain classification, both the intraclass feature cohesion and the interclass feature dispersion have been improved to varying degrees within each domain and across domains. Although L2D [Fig. 8(g)] has great improvement over the aforementioned SDG methods, it is obviously inferior to ERM in terms of *building* on the Satellite II dataset. And evidently, as shown in Fig. 8(n), CCDR can more tightly congregate the features from *building*, and more significantly maximize the distances between these features and those from *background*.

After that, we further discuss the feature visualization results of CCDR and its variants. Compared to ERM, the first three variants without class-aware domain alignment, as shown in Fig. 8(i)–(k), can obviously lower the confusion between features from different classes, but their semantic correlation across domains requires further optimization. In contrast, the last two variants and CCDR, which are organized through adding the cross-domain constraint $\mathcal{L}_{\text{align}}$ to the first three variants, can significantly bring closer features from the same class but different domains, while pushing apart features from different classes across domains, as shown in Fig. 8(l)–(n).

In short, the feature visualization results further verify that CCDR has remarkable advantages over other state-of-the-art SDG methods in eliminating the interference of domain shift in the remote sensing field; the strategy of auxiliary domain generation adopted by CCDR can encourage the segmentation model to aggregate domain-invariant features; and the cross-domain constraint for class-aware domain alignment can effectively boost the semantic correlation across domains.

V. Conclusion

To center on more realistic requirements, this article proposes a novel SDG method for remote sensing image segmentation, termed CCDR, which performs auxiliary domain generation via domain randomization of texture and style information and further ensures the semantic correlation within each domain and across domains through multidomain supervised learning and class-aware domain alignment, respectively. To delve into CCDR, this article conducts extensive experiments on the WHU and OM datasets to evaluate its effectiveness in remote sensing image segmentation, analyze the influence of its hyperparameters, probe the specific effects of each component, assess its scalability in different segmentation models, and further reveal its performance via data and feature visualization. The experimental and visualization results are indicated as follows.

1) In comparison to other state-of-the-art SDG methods, CCDR showcases superior performance in remote sensing image segmentation tasks and does not require a sophisticated training pipeline or cumbersome generation process.
2) The data generation module of CCDR, namely, TSRM, not only does not require additional training but also excels in simulating domain shift in remote sensing, compared with other SDG methods.
3) The cross-domain constraint for class-aware domain alignment in CCDR can effectively improve the reliability of classification in unseen target domains.

However, since TSRM is a completely random process, some generated data may have severe semantic distortions, impairing classification. Although this negative impact has been controlled through residual connection and class-aware domain alignment to a certain extent, the accuracy of *building* from CCDR is still lower than that from ERM in the Pa→Po experiment. Therefore, in the future, we will further investigate how to generate the auxiliary domain that can diversify the domain distributions, while preserving the original semantics as much as possible.

References

[1] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, Mar. 2020.

[2] S. Wang, T. Zhou, Y. Lu, and H. Di, "Contextual transformation network for lightweight remote-sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5615313, doi: 10.1109/TGRS.2021.3132093.

[3] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[4] Z. Zhang et al., "Multireceptive field: An adaptive path aggregation graph neural framework for hyperspectral image classification," *Expert Syst. Appl.*, vol. 217, May 2023, Art. no. 119508.

[5] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858.

[6] Y. Ding et al., "Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5536716, doi: 10.1109/TGRS.2022.3202865.

[7] X. Zhang, C. Liu, and C. Y. Suen, "Towards robust pattern recognition: A review," *Proc. IEEE*, vol. 108, no. 6, pp. 894–922, Jun. 2020.

[8] S. Sankaranarayanan, Y. Balaji, C. D. Castillo, and R. Chellappa, "Generate to adapt: Aligning domains using generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8503–8512.

[9] Y. Li, L. Yuan, and N. Vasconcelos, "Bidirectional learning for domain adaptation of semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6936–6945.

[10] Y. Yang, D. Lao, G. Sundaramoorthi, and S. Soatto, "Phase consistent ecological domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 9011–9020.

[11] C. Liang, B. Cheng, B. Xiao, and Y. Dong, "Unsupervised domain adaptation for remote sensing image segmentation based on adversarial learning and self-training," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[12] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*.

[13] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1180–1189.

[14] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:10:47 UTC from IEEE Xplore. Restrictions apply.

[15] C. Liang, B. Cheng, B. Xiao, Y. Dong, and J. Chen, "Multilevel heterogeneous domain adaptation method for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5601916, doi: 10.1109/TGRS.2023.3236957.

[16] X. Fan, Q. Wang, J. Ke, F. Yang, B. Gong, and M. Zhou, "Adversarially adaptive normalization for single domain generalization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8208–8217.

[17] R. Gong, W. Li, Y. Chen, and L. Van Gool, "DLOW: Domain flow for adaptation and generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2477–2486.

[18] X. Peng, Z. Huang, X. Sun, and K. Saenko, "Domain agnostic learning with disentangled representations," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 5102–5112.

[19] H. Zhang, Y.-F. Zhang, W. Liu, A. Weller, B. Scholkopf, and E. P. Xing, "Towards principled disentanglement for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8024–8034.

[20] H. Qian, S. J. Pan, and C. Miao, "Latent independent excitation for generalizable sensor-based cross-person activity recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 11921–11929.

[21] S. Shankar, V. Piratla, S. Chakrabarti, S. Chaudhuri, P. Jyothi, and S. Sarawagi, "Generalizing across domains via cross-gradient training," 2018, *arXiv:1804.10745*.

[22] D. Li, Y. Yang, Y.-Z. Song, and T. Hospedales, "Learning to generalize: Meta-learning for domain generalization," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 3490–3497.

[23] Y. Balaji, S. Sankaranarayanan, and R. Chellappa, "MetaReg: Towards domain generalization using meta-regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 998–1008.

[24] Y. Li, Y. Yang, W. Zhou, and T. Hospedales, "Feature-critic networks for heterogeneous domain generalization," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 3915–3924.

[25] R. Volpi, H. Namkoong, O. Sener, J. Duchi, V. Murino, and S. Savarese, "Generalizing to unseen domains via adversarial data augmentation," 2018, *arXiv:1805.12018*.

[26] F. Qiao, L. Zhao, and X. Peng, "Learning to learn single domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12556–12565.

[27] L. Zhao, T. Liu, X. Peng, and D. Metaxas, "Maximum-entropy adversarial data augmentation for improved generalization and robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 14435–14447.

[28] L. Li et al., "Progressive domain expansion network for single domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 224–233.

[29] X. Yue, Y. Zhang, S. Zhao, A. Sangiovanni-Vincentelli, K. Keutzer, and B. Gong, "Domain randomization and pyramid consistency: Simulation-to-real generalization without accessing target domain data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2100–2110.

[30] Z. Wang, Y. Luo, R. Qiu, Z. Huang, and M. Baktashmotlagh, "Learning to diversify for single domain generalization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 814–823.

[31] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, "AugMix: A simple data processing method to improve robustness and uncertainty," 2019, *arXiv:1912.02781*.

[32] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang, "Domain generalization with mixstyle," 2021, *arXiv:2104.02008*.

[33] Z. Xu, D. Liu, J. Yang, C. Raffel, and M. Niethammer, "Robust and generalizable visual representation learning via random convolutions," 2020, *arXiv:2007.13003*.

[34] W. Liu, F. Su, X. Jin, H. Li, and R. Qin, "Bispace domain adaptation network for remotely sensed semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5600211, doi: 10.1109/TGRS.2020.3035561.

[35] L. Yan, B. Fan, H. Liu, C. Huo, S. Xiang, and C. Pan, "Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3558–3573, May 2020.

[36] J. Chen, J. Zhu, Y. Guo, G. Sun, Y. Zhang, and M. Deng, "Unsupervised domain adaptation for semantic segmentation of high-resolution remote sensing imagery driven by category-certainty attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616915.

[37] Y. Zhao, S. Li, C. H. Liu, Y. Han, H. Shi, and W. Li, "Domain adaptive remote sensing scene recognition via semantic relationship knowledge transfer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, Art. no. 2001013, doi: 10.1109/TGRS.2023.3267149.

[38] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 1–15.

[39] J. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2223–2232.

[40] T. Kim, M. Cha, H. Kim, J. K. Lee, and J. Kim, "Learning to discover cross-domain relations with generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1857–1865.

[41] Z. Yi, H. Zhang, P. Tan, and M. Gong, "DualGAN: Unsupervised dual learning for image-to-image translation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2849–2857.

[42] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5609413, doi: 10.1109/TGRS.2021.3104032.

[43] B. Benjdira, Y. Bazi, A. Koubaa, and K. Ouni, "Unsupervised domain adaptation using generative adversarial networks for semantic segmentation of aerial images," *Remote Sens.*, vol. 11, no. 11, p. 1369, 2019.

[44] L. Shi, Z. Wang, B. Pan, and Z. Shi, "An end-to-end network for remote sensing imagery semantic segmentation via joint pixel- and representation-level domain adaptation," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1896–1900, Nov. 2021.

[45] O. Tasar, S. L. Happy, Y. Tarabalka, and P. Alliez, "ColorMapGAN: Unsupervised domain adaptation for semantic segmentation using color mapping generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7178–7193, Oct. 2020.

[46] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[47] K. Zhou, Y. Yang, T. Hospedales, and T. Xiang, "Learning to generate novel domains for domain generalization," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 561–578.

[48] Y. Zhao et al., "Learning to generalize unseen domains via memory-based multi-source meta-learning for person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6277–6286.

[49] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[50] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[51] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1501–1510.

[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[53] E. Hoffer and N. Ailon, "Deep metric learning using triplet network," in *Proc. Int. Workshop Similarity-Based Pattern Recognit. (SIMBAD)*, Copenhagen, Denmark. Cham, Switzerland: Springer, 2015, pp. 84–92.

[54] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[55] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.

[56] P. Kaiser, J. D. Wegner, A. Lucchi, M. Jaggi, T. Hofmann, and K. Schindler, "Learning aerial image segmentation from online maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6054–6068, Nov. 2017.

[57] ISPRS. *2D Semantic Labeling Contest*. Accessed: Nov. 8, 2021. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/semantic-labeling.aspx

[58] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," *IEEE Pervasive Comput.*, vol. 7, no. 4, pp. 12–18, Oct. 2008.

[59] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," 2017, *arXiv:1706.08500*.

[60] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 2234–2242.

[61] T. Kynkäänniemi, T. Karras, S. Laine, J. Lehtinen, and T. Aila, "Improved precision and recall metric for assessing generative models," 2019, *arXiv:1904.06991*.

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[62] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.
[63] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
[64] V. Koltchinskii, *Oracle Inequalities in Empirical Risk Minimization and Sparse Recovery Problems: École D'Été de Probabilités de Saint-Flour XXXVIII-2008*, vol. 2033. Cham, Switzerland: Springer, 2011.
[65] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.



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